

Sam Goodwin

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Profile

Computer Science graduate (Bridgewater State University, 2025, Magna Cum Laude) with hands-on experience building cross platform software in Golang, Python, Java and C/C++. Projects span desktop applications, concurrent systems, computer vision and IoT, each built with an emphasis on clean architecture, thorough testing and clear documentation. Certified by HackerRank in SQL, Go, Java, and Python.

Education

Bachelor of Science in Computer Science

Bridgewater State University, 2025

- Magna Cum Laude (GPA: 3.8)
- Dean's List (All Semesters)

Skills

Languages: Golang, Python, Java, C++, Bash/Shell

Databases: SQL and SQLite

Libraries/Frameworks: OpenCV, DJI Tello, Excelize, SQLite3, Fyne, JavaFX, Maven, face_recognition, MediaPipe

Tools: Git, GitHub, Arduino, Docker, Raspberry Pi, VS Code, PlatformIO, JetBrains

Operating Systems: Linux and Windows

Other: GUI/CLI development, Computer vision, Data Visualization, Mapping, IoT Systems, API integration

Certifications

- SQL (Advanced, Intermediate, Basic) Skills Certification – HackerRank
- Go (Intermediate and Basic) Skills Certification – HackerRank
- Java (Basic) Skills Certification – HackerRank
- Python (Basic) Skills Certification – HackerRank

Projects

Job Visualizer

GitHub: github.com/goodwin-sam/job-visualizer

- Cross-platform Go desktop app (GUI + CLI) for importing, processing, and visualizing job data from Excel files, backed by a SQLite database.
- Built interactive GUI with Fyne for file filtering by keyword, location, salary, and remote status, plus browser-based mapping via OpenStreetMap Nominatim API with local geocoding cache, and interactive charts for data visualization.
- Enforced clean architecture through cmd/pkg separation, conventional commits, comprehensive testing, detailed documentation, and portable release binaries for Linux and Windows.

Concurrent File Processor

GitHub: github.com/goodwin-sam/concurrent-file-processor

- Cross-platform Java desktop app for concurrent multi-file text analysis, computing word frequencies, character/line counts, and aggregate file statistics via GUI or headless CLI.
- Implemented custom work-stealing thread pools for load-balanced parallel processing, with a JavaFX multi-window GUI and portable executables for Windows and Linux.
- Designed thread-safe modular architecture with clear package separation, comprehensive unit tests, and thorough error handling.

Smart Tello Follower

GitHub: github.com/goodwin-sam/smart-tello-follower

- Python computer vision system that streams live video from a DJI Tello drone, performs real-time face recognition using OpenCV and the face_recognition library to autonomously track a target.
- Implemented gesture recognition via MediaPipe to allow hand-signal-based flight control, with modular separation of drone control, face recognition, and gesture recognition.
- Designed configurable settings via a dedicated config module, with clean dependency management.

Environment Sensor Hub

GitHub: github.com/goodwin-sam/enviro-sensor-hub

- Arduino-based C++ environmental monitoring system that reads temperature, humidity, water level, and ambient light from multiple sensors, driving a fan, buzzer, and 16x2 LCD in response.
- Implemented PWM fan speed control across three temperature tiers, automatic LCD backlight adjustment based on ambient light, and configurable buzzer alarms for threshold violations.
- Structured into a modular package structure (sensors, display, serial utilities) with all thresholds and pin assignments centralized in a config header for easy tuning.

Experience

Property Manager

Private Landlord, Massachusetts

Summer 2020 – Present

- Managed day to day operations for multiple residential rental properties, responding to landlord directed tenant requests often same day or next day. Including on-call night/weekend work for critical issues such as no heat or no hot water.
- Performed hands on repairs, lawn care, snow removal and general maintenance personally, troubleshooting and/or resolving plumbing, electrical, appliance, and other general issues.
- Collected and processed rental payments while maintaining accurate records and handling administrative tasks.
- Scheduled and oversaw external contractors when needed, ensuring timely completion and quality work.
- Handled documentation, scheduling, and follow-ups, demonstrating strong organization and reliability in a fast-paced, high demand environment.

Teaching Assistant/Tutor

Curry College, Milton Massachusetts

January 2019 – May 2020

- Delivered one-on-one and small-group tutoring sessions for undergraduate students in Chemistry and Biology courses.
- Assisted students in understanding complex scientific material, breaking down problems, explaining experimental processes, and improving critical thinking and problem-solving skills.
- Held regular office hours and review sessions to resolve conceptual misunderstandings, reviewing homework/lab reports, and provide constructive feedback to enhance comprehension and academic performance.
- Supported faculty by preparing study materials, facilitating group discussions, and helping students navigate rigorous science coursework